

Laiba Mazhar

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EDUCATION

FAST National University of Computer and Emerging Sciences

Aug 2022 – May 2026

BS Data Science | Islamabad, Pakistan | CGPA 3.12 / 4.00

- Final year project: **NexusPilot** — an integrated business platform unifying finance, HR and CRM workflows over a 15-entity relational schema.
- Relevant coursework: Machine Learning, Data Mining, Artificial Intelligence, Parallel and Distributed Computing, Database Systems, Statistics, Data Warehousing, Computer Networks.
- Higher Secondary (FSc Pre-Engineering)**, Aspire College, 2019–2021 — 1024/1100 (93.09%), Grade A+, 2nd position in college, fully funded scholarship.
- Secondary (Matriculation)**, Government Girls High School, Umer Block, Lahore, 2017–2019 — 1010/1100 (91.82%), 1st position in class, fully funded scholarship.

RESEARCH

Six research entries — five manuscripts, none currently published, and one in-progress industrial R&D record. Status is stated explicitly for each.

Anchored Counterfactual Attribution of Model Degradation to ETL Pipeline Operators

2026

First author, with M. Sarfraz | Manuscript in preparation | github.com/laiba-mazhar/culpa-pipeline-attribution

- Abstract.** When a production model degrades between two scheduled pipeline runs with no code deployed, engineers must determine which operator is responsible. We formulate this as a cooperative game over *hybrid replays* of a fixed pipeline across two temporal snapshots, whose Shapley value decomposes the observed degradation exactly across operators. We show the standard Shapley value is correct as accounting but poor as triage, and introduce the **incident-anchored Shapley value**, which recovers both standard Shapley and leave-one-out ablation as boundary cases. On real NYC taxi partitions it reaches 100% top-1 accuracy where leave-one-out drops to 50%, while retaining exact decomposition to 3×10^{-17} . Output-hash pruning reduces 2^{33} hybrid replays to three model fits.
- Open artifact with eight reproducible experiments, full run logs and an IEEE-format manuscript. Also establishes that current stagewise practice is order-dependent: attributed blame reverses sign across valid topological orders.

Decoupling Material Recognition from Disposition Policy in Edge Vision Waste Attribution

2026

Industrial R&D at AiPixal | in progress | technical record supporting an HMRC R&D relief claim

- Problem.** An edge vision system on a Jetson Orin NX must identify raw material in a commercial kitchen and separate usable stock from waste, writing an itemised entry to an immutable ledger. Training a detector directly on *useful* and *waste* cannot work: those categories describe provenance and intent rather than appearance, so a network asked to infer them from pixels approximates the rule using whatever correlates in the training set — background, container, capture session — and fails outside it.
- Design.** Two label systems held apart: a detector over 32 concrete material classes, and a disposition mapping to useful / waste / packaging / review that is per-site configuration rather than something learned, with asymmetric per-class thresholds and a review route that never auto-commits. Reclassifying an ingredient becomes one line of YAML.
- Method.** The dataset programme is built as gates that fail the build rather than checks that report, because the three failure modes that matter — silent ingest divergence, near-duplicate leakage across splits, and class identity confounded with capture session — all inflate reported accuracy rather than depress it, so none announces itself. Includes a written annotation protocol and an inter-annotator agreement procedure.

Safety-First Maternal Risk Screening: Guideline Calibration and Urdu Risk Communication

2026

Sole author | R&D complete, IEEE-format manuscript | github.com/laiba-mazhar/maternal-health-risk

- Abstract.** Published work on the UCI Maternal Health Risk dataset converges on one shape: train a tree ensemble, report 80–85% accuracy, conclude that ML is promising for maternal healthcare. This R&D project asks a different question — whether a prediction changes the referral decision, and whether a Lady Health Worker can say it out loud. Accuracy is replaced by a **critical-miss rate**, a cost matrix and a fixed referral budget, so sending a high-risk mother home is scored differently in kind from escalating her unnecessarily. Because the data was collected in Bangladesh and its labelling protocol was never published, eleven sourced clinical thresholds serve as an independent baseline rather than assuming label transfer.
- Output is delivered as hand-authored bilingual Urdu risk bands with SHAP attributions, not a class index. 155 passing tests and a nine-page IEEE-format manuscript. A research prototype, explicitly not a medical device.

Adversarial Data Poisoning Detection in Federated Learning via ICS

2025

Sole author | Manuscript in preparation

- **Abstract.** Proposes the *Independent Class-wise Coherence Score* (ICS), a defense against label-flipping and backdoor attacks in federated learning, using MI-FGSM adversarial probing and four fused coherence metrics. Across 15 distributed clients on MNIST, EMNIST and CIFAR-10, ICS outperforms global-coherence baselines by 5–7% accuracy and reduces attack success rate by 30–40% (75.02% accuracy at 19.75% ASR on MNIST).

Robust and Explainable Multiclass Intrusion Detection with XGBoost and SHAP

2025

Sole author | Manuscript in preparation

- **Abstract.** A statistically validated, explainable multiclass intrusion detection system combining XGBoost, TreeSHAP and a severity scoring engine for SOC deployment, evaluated on NSL-KDD and UNSW-NB15. Achieves 99.90% average accuracy, a 12.28% macro-F1 gain over a Random Forest baseline and a 2.44× stability improvement across five seeds, emitting JSON-exportable SOC alert packets.

LLM-Driven Autonomous Trading Agents with Hallucination Mitigation

2025

Sole author | Manuscript in preparation

- **Abstract.** A three-agent pipeline (Analyst, Critic, Decision) built on Llama-3.3-70B to mitigate hallucination in financial reasoning over real-time market data. The Critic agent emits structured JSON verification reports, achieving an 80% recommendation adjustment rate and consistent hallucination detection across five major US equities.

WORK EXPERIENCE

AIpIXal — Johar Town, Lahore, Pakistan (On-site)

Aug 2026 – Present

Senior ML Automation Engineer | Full-time

- Edge vision for material recognition on an NVIDIA Jetson Orin NX: identifying raw material in a commercial kitchen and separating usable stock from waste for an itemised, immutable ledger. Own the annotation programme and the label design — a 32-class material taxonomy held apart from a per-site disposition policy, so reclassifying an ingredient is a configuration change rather than a retrain.
- Built the dataset integrity gates as build-failing steps rather than reports: ingest reconciliation, group-aware splitting, class-versus-session confound auditing and a measured minimum resolvable fragment size. Work is documented as the technical record supporting an HMRC R&D relief claim.

CloudWorks — Texas, USA (Remote)

Jun 2026 – Present

Data Developer and Analyst | Part-time

- Build and maintain data pipelines and analytics for US healthcare clients, including CareFlow, a care-coordination platform with role-based access control, HIPAA-compliant data handling and full audit logging.

AI Muttaqeen Institute — Remote

Nov 2025 – May 2026

Database Manager | Full-time

- Built automated validation, reporting and reconciliation pipelines for institutional records, improving cross-departmental data accuracy and audit readiness.

Tashi Technologies Corp — Islamabad, Pakistan

Jul 2025 – Oct 2025

Data Automation Engineering Intern | On-site

- Built ETL workflows and classification/regression pipelines (XGBoost, scikit-learn); integrated AI APIs for process automation and deployed models to cloud infrastructure for real-time scoring.

Independent / Freelance — Remote

Jun 2023 – Jun 2025

Data and AI Engineer, international clients

- Delivered batch and streaming pipelines (Spark, Kafka, Airflow) on GCP and AWS feeding ML models and BI dashboards; built LLM prototypes with RAG and agentic workflows.

SELECTED TECHNICAL PROJECTS

Research projects appear under Research above and the final year project under Education; these are the remainder. Code for all of them is public.

Machine learning, NLP and agents

- **English-Urdu Neural Machine Translation** — Transformer built from scratch and benchmarked against an LSTM/Bahdanau-attention seq2seq baseline on the UMC005 corpus; BPE tokenization, BLEU/ROUGE evaluation.
- **ragforge** — hybrid retrieval combining dense embeddings, BM25 and reciprocal rank fusion, with Chroma and Qdrant backends, parameter sweeps and a CI regression gate, so a RAG configuration is measured rather than guessed at.
- **LaunchMind** — multi-agent system that runs a micro-startup end to end with no human in the loop: idea through to GitHub pull request, launch post and cold outreach.
- **Bagging vs Boosting** — comparative study of decision trees, bagging, AdaBoost and random forests on breast-cancer classification, with the ensemble methods implemented from scratch rather than called from a library.

Data engineering and analytics

- **Airflow E-Commerce Pipeline** — end-to-end Apache Airflow pipeline over the Brazilian e-commerce dataset: separate extraction, preprocessing and analysis DAGs producing scheduled visual reports.
- **Real-Time Telecom Network Analytics** — Kafka/Spark/Airflow streaming pipeline with CI/CD and DAG version control via GitHub Actions.

- **E-Commerce Retail Trends and CLV** — effect of global e-commerce on traditional retail, with customer-lifetime-value modelling and what-if pricing analyses.
- **Iran Airlines Environmental Analysis** — Tableau study of temperature, pollutant and wind-speed trends across nine sheets and three dashboards.

Systems and full-stack

- **Maktab** — academy management system covering student and staff records, timetabling, attendance, fees and exams on Supabase Postgres with row-level security.
- **AI Handwriting Analysis and Author Identification** — Flask application combining OCR, ML and NLP for author identification, sentiment and emotion analysis and grammar checking.
- **Parallel and Distributed Computing** — Pthreads, OpenMP and OpenCL implementations: matrix operations, CPU-affinity tuning, Game of Life, box blur and GPU convolution.
- **COAL Taxi Dash** — console driving game written entirely in 32-bit MASM assembly, with procedural map generation, collision-based scoring and file-backed persistence.

TECHNICAL SKILLS

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, pandas, NumPy, SciPy, federated learning, adversarial ML, explainable AI (SHAP), cost-sensitive learning, NLP, time-series forecasting, LLM agents, RAG.
- **Data engineering:** Apache Spark, PySpark, Kafka, Airflow, dbt, Databricks, Snowflake, BigQuery, ETL/ELT design.
- **Programming:** Python, SQL, TypeScript/Javascript, C/C++, Java (basic), MASM assembly.
- **Infrastructure:** GCP, AWS, Docker, GitHub Actions, PostgreSQL, MongoDB, Supabase, Chroma, Qdrant, FastAPI, Django REST, row-level security, audit logging, HIPAA-compliant handling.
- **Analysis and visualization:** Power BI, Tableau, Matplotlib, Seaborn, Streamlit, LaTeX, pytest.

TRAINING AND CERTIFICATIONS

- **Machine Learning** — Stanford University, via Coursera (Andrew Ng).
- **Virtual internships in Python, C++ and Java programming** — CodSoft.
- **Relevant university coursework** — Machine Learning, Data Mining, Artificial Intelligence, Generative AI, Parallel and Distributed Computing, Database Systems, Data Warehousing, Statistics, Computer Networks.

AWARDS AND HONOURS

- **Winner, Big Data Quest Hackathon** — DataFest-25, NaSCon: predictive analytics solution built on PySpark and cloud infrastructure.
- **Speed Programming Certificate** — NaSCon-24.
- **Certificates of Special Mention**, Model United Nations — CUIMUN (COMSATS), 2023; BUIMUN (Islamabad), 2024.
- **Winner**, university arts competition — FAST-NUCES.
- **Winner, Best Guerrilla Marketing** (cash prize) — digital marketing course project for the Starbucks brand.
- Academic distinction shields at secondary and higher-secondary level.

EXTRACURRICULAR ACTIVITIES AND INTERESTS

Rah-e-Haq (NGO) — Islamabad Nov 2023 – Apr 2025
Donation Officer, volunteer

- Maintained structured donation and financial records; coordinated cross-team deliverables and transaction accountability.

Baitulnoor — Pakistan 1.5 years
Head of Investigations, volunteer

- Led verification and investigation processes across cases; documented findings into actionable stakeholder reports.
- **Founder, Datahelyx** — a data and AI consultancy (datahelyx.com), currently building its first client engagements.
- **Founder and artist, Rangrayze** — a gallery for Sufi-inspired calligraphy and canvas work; manages creative production, commissions and digital marketing.
- **Founder, airRTH** — a community climate initiative running sustainability awareness and environmental action programs, at an early stage.
- **Model United Nations** — delegate across multiple national conferences; two Special Mention awards.
- **Sport** — three medals in inter-college competition.
- **Interests** — trustworthy and diagnosable ML systems, data engineering for AI, distributed systems, calligraphy, climate advocacy, competitive debate.

LANGUAGES

English (fluent, professional working proficiency) | Urdu (native) | German (basic) | Arabic (basic reading and writing)